

StructDiff: A Training-Free Framework for Precise Structural Control in Remote Sensing Image Synthesis

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Abstract—Diffusion models have advanced remote sensing (RS) image synthesis, yet maintaining strict geospatial integrity remains difficult. Existing methods face a practical trade-off: training-based approaches are condition-specific and require costly retraining, whereas conventional training-free techniques often struggle to balance structural preservation with prompt-driven appearance control. In this paper, we present StructDiff, a training-free framework for precise structural control in RS image synthesis without task-specific retraining for new control conditions. Its core component, the Fixed-Timestep Structural Conditioner (FTSC), extracts a geometric prior from a single fixed diffusion timestep and uses it as a stable structural constraint during generation. StructDiff is further equipped with a Multimodal Prompt Modulator (MPM), implemented with a frozen off-the-shelf vision-language model, and a Restart Refinement Strategy (RRS) for late-stage artifact suppression. Experiments on DIOR and DOTA, together with a downstream object detection study under low-data regimes, show that StructDiff provides a strong balance between geospatial accuracy, semantic alignment, and perceptual quality among training-free controllable baselines, supporting its use in controlled augmentation workflows such as on-the-fly data augmentation and rapid virtual prototyping in remote sensing. Code is publicly available at: <https://github.com/bxuanz/StructDiff>.

Index Terms—Remote Sensing, Diffusion Models, Structural Control, Training-Free, Image Synthesis

I. INTRODUCTION

High-quality remote sensing (RS) images are [1] indispensable for environmental monitoring, urban planning, and disaster assessment. The development and validation of Earth observation models [2–6]—from semantic segmentation to change detection—depend heavily on such data. However, RS image collection is often limited by long revisit cycles, adverse atmospheric conditions, and high acquisition costs. This scarcity has motivated image synthesis methods, with diffusion models [7–11] emerging as a powerful paradigm for data augmentation and virtual simulation.

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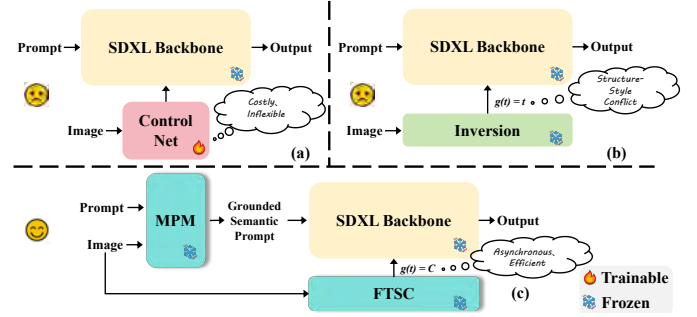


Fig. 1. Conceptual comparison of controllable generation paradigms. (a) The *training-based* approach typically requires costly condition-specific retraining. (b) Conventional *training-free* methods using synchronous injection ($g(t) = t$) struggle to balance structure and prompt alignment. (c) Our proposed framework, **StructDiff**, alleviates these limitations through a fixed-timestep conditioner (FTSC), prompt modulation (MPM), and a late-stage refinement step (RRS).

Despite their success in general image synthesis [9, 10, 12], diffusion models face a fundamental RS-specific challenge: balancing prompt-driven appearance with strict structure preservation. Over-emphasizing appearance can degrade or hallucinate key geographic features such as roads, buildings, and rivers, whereas prioritizing structure can weaken thematic alignment. This tension hinders high-precision RS synthesis [6, 13–15]. We refer to it as the *structure-prompt conflict*. For synchronous feature-injection pipelines, this task-level conflict is driven by an *integrity-leakage dilemma*: later low-noise features preserve finer structure but carry more condition-domain artifacts, whereas earlier high-noise features are cleaner but too coarse to maintain detailed spatial integrity.

Mainstream approaches fall into two paradigms. The first is *training-based*, as illustrated in Fig. 1(a). Methods such as ControlNet [16] require dedicated condition adapters trained on paired data, so each structural control family needs its own fine-tuning pipeline and evaluation setting. This retraining burden motivates more flexible *training-free* solutions. However, existing training-free approaches, shown in Fig. 1(b), often manipulate internal features [17, 18] or attention maps [19] through synchronous injection ($g(t) = t$), which makes it difficult to balance geospatial preservation with prompt-driven appearance rendering in RS imagery.

To address these limitations, we propose StructDiff, a training-free framework for RS image synthesis. Throughout this paper, “training-free” means that adapting to a new control type requires no task-specific or condition-specific

optimization, although StructDiff still relies on frozen pre-trained foundation models at inference time. Its core is the Fixed-Timestep Structural Conditioner (FTSC), which extracts a structural prior from a single intermediate diffusion step C . This *sweet spot* preserves geospatial layout before features become either overly contaminated by modality-specific details or too coarse to maintain local structure. By anchoring generation to this stable prior ($g(t) = C$), FTSC provides structural guidance throughout the controlled stage while leaving room for prompt-guided appearance rendering. In this framing, FTSC is the structural-control backbone of StructDiff rather than a standalone endpoint for final image quality; the final performance is achieved through its coordination with the semantic and refinement modules described below.

On top of this structural backbone, our ablation analysis reveals two complementary modules. First, concise user prompts often under-specify RS scenes, so we introduce the Multimodal Prompt Modulator (MPM), which uses a frozen VLM such as LLaVA [20] to convert the structural condition and raw prompt into a grounded semantic prompt. Second, residual low-noise artifacts may remain after the structure-controlled stage, so we incorporate a Restart Refinement Strategy (RRS), inspired by restart-style diffusion refinement [21, 22], as a lightweight polishing stage.

In summary, the primary contributions of this work are as follows:

- 1) We present StructDiff, a training-free framework for flexible structural control in RS image synthesis without task-specific retraining for new condition types.
- 2) We introduce FTSC as the structural-control backbone, which mitigates the integrity-leakage dilemma by caching a balanced structural prior from a fixed timestep and reusing it throughout the controlled denoising stage.
- 3) We design two performance-critical complementary modules for RS synthesis: MPM for grounded semantic prompting and RRS for low-noise artifact refinement.
- 4) Experiments on DIOR and DOTA, together with human evaluation and downstream detection analysis, show that StructDiff offers a strong overall balance between geospatial accuracy, semantic alignment, and perceptual quality among training-free controllable baselines.

II. RELATED WORK

This section critically reviews prior work on generative models, structural control (both *training-based* and *training-free*), and large multimodal models to contextualize our framework.

A. Training-Based Methods for Structural Control

A primary approach to enforce precise spatial fidelity is through training-based conditioning. This paradigm involves augmenting a large pretrained model with a dedicated, trainable module designed to interpret specific structural inputs, conditioned on large-scale paired datasets. The seminal work in this area is ControlNet [16], which injects spatial conditions via a trainable copy of the model’s U-Net [23] encoder. These copies are connected through specialized “zero-convolution”

layers that are initialized to have no effect, ensuring that the vast knowledge of the pretrained model is preserved during the initial stages of fine-tuning. T2I-Adapters [24] offer a more parameter-efficient alternative, using small, lightweight modules to inject conditional guidance at multiple feature scales. This paradigm has also been extended to other forms of guidance, such as grounded generation from bounding box layouts.

B. Training-Free Methods for Structural Control

As an alternative to the operational rigidity of training-based models, training-free methods offer a flexible paradigm for controlling generation at inference time without altering model weights. These techniques skillfully manipulate the internal dynamics of a frozen, pretrained model. Broadly, they fall into several categories. One line focuses on modifying attention maps; methods like Prompt-to-Prompt [19] alter cross-attention or self-attention to control semantics or spatial layout. Another major category involves direct feature injection. Early work like Plug-and-Play (PnP) [17] replaced self-attention features with those from a reference image. More recent methods, such as Ctrl-X [25] and FreeControl [26], directly inject features derived from the noisy latent representation of the *structural condition* (e.g., edge maps, depth maps) into the generation process at each step.

Despite their flexibility, a majority of these feature injection methods rely on a synchronous strategy ($g(t) = t$), where guidance is applied continuously and identically across denoising steps. This creates a fundamental conflict between structural fidelity and domain alignment, particularly problematic for high-precision RS tasks. The diffusion process operates on a coarse-to-fine trajectory: early steps define global composition, while later steps refine high-frequency details. Applying a consistent structural constraint derived from, say, a grayscale depth map during the texture-refinement phase often leads to critical failure modes in RS image synthesis. First, *style contamination or leakage* occurs, where irrelevant attributes from the condition (e.g., the grayscale nature of edge maps, sensor noise patterns) “bleed” into the generated RGB image, compromising its photorealism and domain authenticity [27]. Furthermore, *structural decay* becomes evident as the injected structural signal is progressively overpowered by the model’s strong internal priors for natural images, causing an erosion of fine-grained geospatial details (e.g., blurring of building footprints or road networks) crucial for RS applications. This unresolved tension limits the practical effectiveness of existing training-free methods for tasks demanding high geospatial accuracy, motivating our work to develop a conditioning strategy that explicitly resolves this core conflict.

C. Multimodal Models for Semantic Refinement

Beyond geometric control, achieving fine-grained semantic and stylistic fidelity is crucial for generating realistic and contextually appropriate remote sensing (RS) imagery. Historically, this aspect of synthesis was an artisanal process, relying heavily on manual prompt engineering. Users engaged in trial-and-error discovery of keywords and phrases to coax

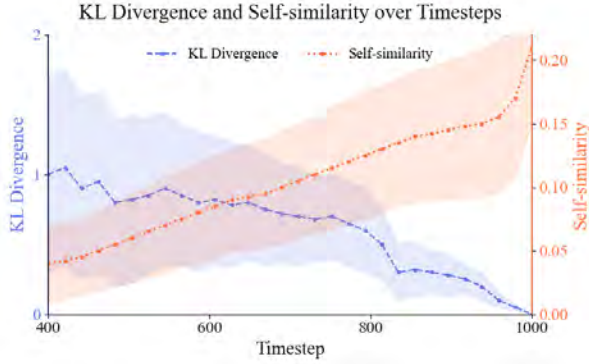


Fig. 2. Empirical analysis of the integrity-leakage dilemma in synchronous injection methods ($g(t) = t$). As t gets smaller ($1000 \rightarrow 400$), Self-similarity distance decreases while KL divergence sharply increases, showing that structurally informative features are also more prone to leakage.

the desired output, an approach often inadequate for describing the complex spatial arrangements and diverse features typical of RS scenes, and fundamentally lacking the reproducibility required for scientific applications.

A significant paradigm shift has been the advent of Large Multimodal Models (LMMs), which bridge the gap between vision and language by connecting a powerful vision encoder to a large language model [28]. This architecture enables a sophisticated, nuanced understanding of cross-modal concepts, allowing the models to perform complex visual reasoning. This has led to a proliferation of powerful LMMs, from influential open-source models like LLaVA [20], and Qwen-VL [29]. These models excel at “image-to-text” translation [11], articulating intricate visual information in a rich linguistic format that is highly interpretable by text-to-image diffusion models.

This capability suggests a promising avenue for automating and enhancing prompt generation. By leveraging an LMM to translate visual conditions into rich, descriptive text, it may be possible to move beyond manual engineering. However, effectively operationalizing these models to act as “scene interpreters” that fuse user intent with visual data—particularly for specialized domains like RS—remains an open research direction.

III. METHODOLOGY

This section details StructDiff, our training-free framework for high-fidelity structurally guided image synthesis.

Notation. We use t for the current denoising timestep, C for the fixed timestep from which structural priors are cached, τ for the timestep threshold below which structural replacement is disabled, and l for a selected U-Net layer. Throughout this paper, larger t corresponds to an earlier and noisier stage, while smaller t corresponds to a later and cleaner stage. Superscripts *struct*, *app*, and *out* denote the structural-condition branch, the prompt-guided appearance branch, and the main output branch, respectively. Accordingly, $f_{l,C}^{struct}$ and $A_{l,C}^{struct}$ are fixed structural priors, $f_{l,t}^{app}$ is the appearance feature at step t , and $f_{l,t}^{out}$ is the current feature of the main branch after FTSC replacement.

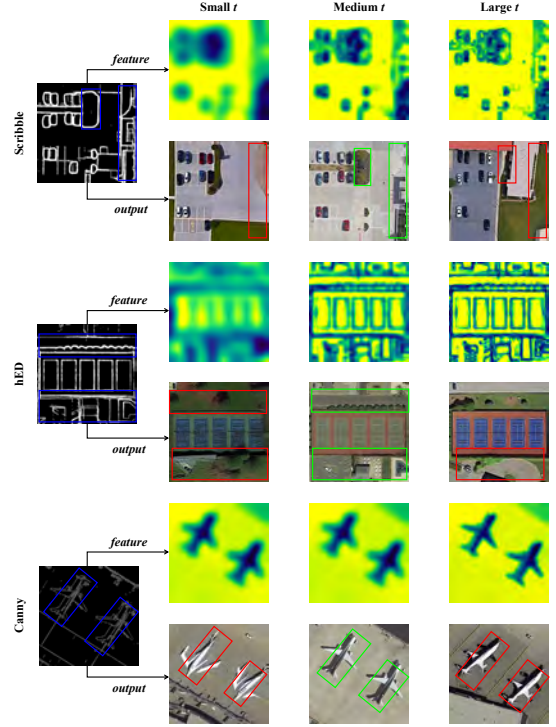


Fig. 3. Qualitative validation of the optimal injection timestep C for FTSC. For each condition type, we visualize large- t , medium- t , and small- t features and the corresponding outputs generated under fixed conditioning ($g(t) = C$). Large- t features miss local detail, small- t features cause *structural misalignments* (red dashed boxes), and medium- t features yield the best structural fidelity (green solid boxes). The full visualization protocol is given in the supplementary material.

A. Motivation: The Integrity-Leakage Dilemma

The broader challenge in controllable RS synthesis is the structure-prompt conflict: the model should preserve the spatial organization implied by I^{struct} while still rendering the thematic appearance requested by $\mathcal{P}$. For synchronous feature-injection methods, this high-level conflict is driven by a lower-level integrity-leakage dilemma. Conventional training-free methods using synchronous injection ($g(t) = t$) face a conflict between structural integrity and condition leakage. Figure 2 quantifies this as a *scissors dilemma*: timesteps with the best structural detail (low Self-similarity as t decreases) simultaneously exhibit the highest feature discrepancy (spiking KL divergence). This proves that the most structurally informative features are also the most “polluting.”

We test this finding qualitatively in Fig. 3 by using features from three fixed stages ($g(t) = C$) as the conditioning prior. For reproducibility, the protocol fixes the layer, aggregation rule, normalization, prompt, seed, and sampler, changing only the cached timestep C . Features from *large* t are too noisy and capture only the coarse layout. Features from *small* t , despite their detail, suffer from the identified pollution (high KL) and cause significant *structural misalignments* (red boxes). Only the *medium-t* *sweet spot* balances detail and purity and consistently produces high structural fidelity (green boxes).

This combined analysis confirms that the optimal structural prior is not dynamic ($g(t) = t$) but static, located in the medium-timestep *sweet spot* ($g(t) = C$). Our FTSC module

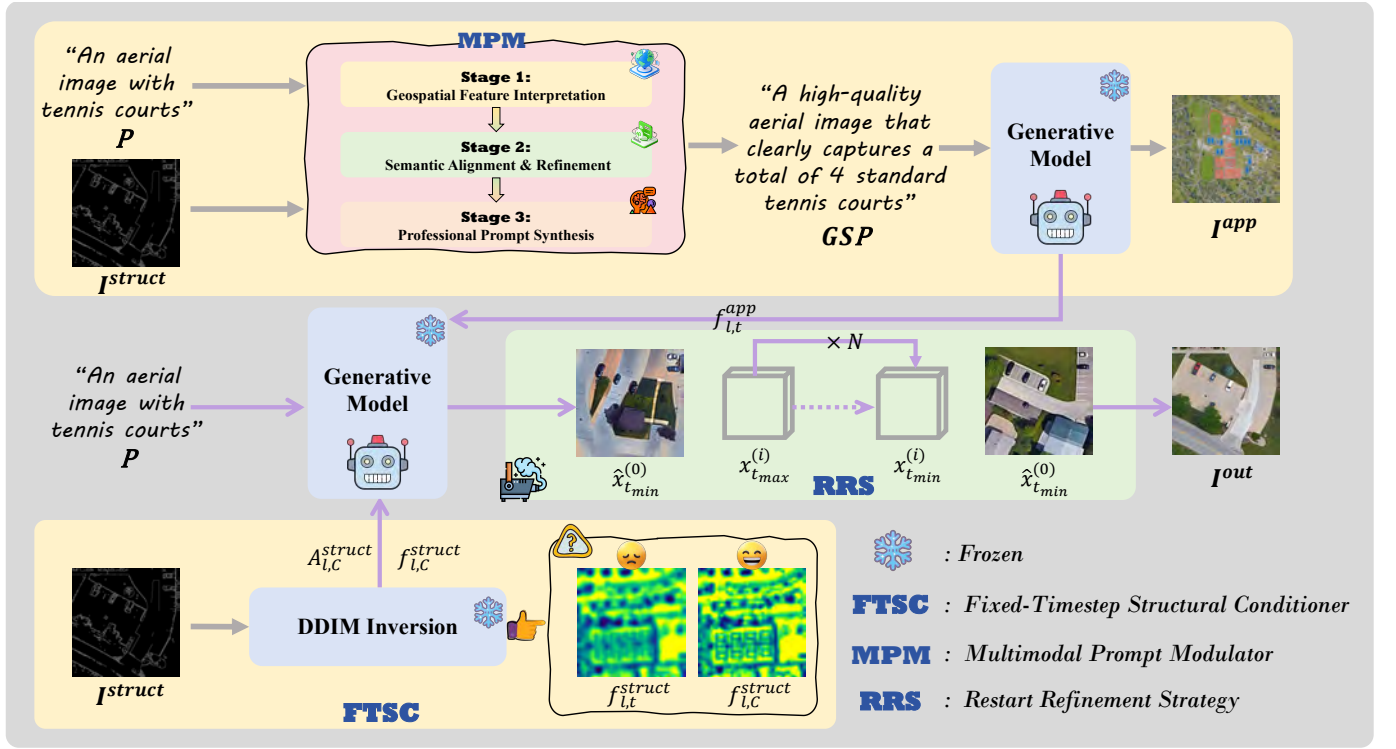


Fig. 4. Overview of **StructDiff**. FTSC caches structural features at timestep C and constrains the main branch, MPM produces the grounded semantic prompt and appearance branch, and RRS performs low-noise refinement. Equation-level variables are defined in the adjacent methodology text.

is designed precisely to exploit this finding.

B. Fixed-Timestep Structural Conditioner

To ensure that the generated image I^{out} adheres to the input structure condition I^{struct} without compromising photorealism, we introduce the Fixed-Timestep Structural Conditioner (FTSC). FTSC serves as the structural-control backbone of the framework, while MPM and RRS provide complementary semantic grounding and low-noise refinement. FTSC addresses the conflict found in prior feature-injection techniques through two coordinated processes: asynchronous structural injection and spatially attuned appearance blending.

1) *Asynchronous Injection for Structural Integrity*: Generally, structural guidance can be parameterized by an injection schedule, denoted $g(t)$, which dictates the source timestep for condition features corresponding to the current generation step t . Many existing methods adopt a synchronous schedule, $g(t) = t$. However, this approach faces a dilemma: features from early denoising steps (t large, high noise) lack sufficient detail, while features from late denoising steps (t small, low noise) often carry strong source-domain characteristics that can contaminate the generation. To circumvent this, FTSC employs an asynchronous, constant injection schedule, $g(t) = C$, where C represents a single predetermined timestep. This decoupling is fundamental, enabling the selection of an optimal feature source independently of the current denoising state t .

Based on our analysis (Sec. III-A), we set the constant timestep C to exploit this intermediate sweet spot ($C = 600$). This mid-range acts as an effective trade-off point, where

features derived from the condition image I^{struct} possess well-defined structural information yet remain noisy enough to align better with the evolving latent states during generation. To obtain these structural representations, we apply DDIM inversion [30] to I^{struct} once up to step C and store the resulting U-Net layer activations, namely the feature maps $f_{l,C}^{struct}$ and self-attention maps $A_{l,C}^{struct}$, for each relevant layer l .

These stored representations function as a static structural template. During the main generation process, we overwrite the corresponding activations in the generative U-Net at each step t where control is active ($t > \tau$). Here, τ is a hyperparameter defining the timestep threshold below which structural guidance is deactivated, allowing finer detail refinement in the late generation stages. Specifically, for each layer l , after computing the native activations $f_{l,t}^{out}$ and $A_{l,t}^{out}$, we perform

$$f_{l,t}^{out} \leftarrow f_{l,C}^{struct}, \quad (1)$$

$$A_{l,t}^{out} \leftarrow A_{l,C}^{struct}. \quad (2)$$

Equation 2 replaces the current main-branch feature and self-attention maps at step t with the cached priors from timestep C , so the layout is repeatedly reset to the same condition-derived structural template while control remains active. This forceful replacement compels the generation process to conform spatially to the cached structural blueprint derived from I^{struct} .

This asynchronous constant injection offers dual benefits. The unwavering structural prior ensures high geometric fidelity throughout the controlled generation phase ($t > \tau$). Moreover, caching the structural representations after a single compu-

tation significantly reduces latency compared to synchronous methods that require reprocessing the condition at every step t .

2) *Spatially-Attuned Appearance Blending*: Within the main generative model, once structural integrity is established at step t via the FTSC replacement (Eq. 2), the features $f_{l,t}^{out}$ are ready to receive the target appearance. This is achieved using appearance features $f_{l,t}^{app}$, sourced from the parallel generation branch guided by the Grounded Semantic Prompt (GSP) (see Fig. 4). Importantly, FTSC does not directly modify the prompt. Instead, it constrains the spatial carrier $f_{l,t}^{out}$ of the main branch, while the prompt-guided appearance branch supplies $f_{l,t}^{app}$. This separation explains how fixed structural guidance can coexist with prompt-driven appearance control. To ensure that textures and colors are applied coherently based on spatial correspondence, we employ a cross-attention mechanism. First, we compute attention weights between the structurally defined output features and the appearance features:

$$A_{cross} = \text{Softmax} \left(\frac{Q_{out} K_{app}^T}{\sqrt{d_k}} \right). \quad (3)$$

Equation (2) defines A_{cross} as the spatial correspondence map: larger entries mean that a location in $f_{l,t}^{out}$ should borrow appearance information more strongly from the matching location in $f_{l,t}^{app}$. Here, the query Q_{out} originates from the structurally correct output features $f_{l,t}^{out}$, and the key K_{app} originates from the appearance features $f_{l,t}^{app}$. They are computed as

$$Q_{out} = \text{Norm}(f_{l,t}^{out}) W_l^Q, \quad (4)$$

$$K_{app} = \text{Norm}(f_{l,t}^{app}) W_l^K. \quad (5)$$

Normalization suppresses raw scale differences before attention, so the weights focus on spatial compatibility.

In Fig. 4, this step corresponds to the bridge between the structurally constrained main branch and the prompt-guided appearance branch. A_{cross} is therefore not a separate generation stream; it is the correspondence map used to read appearance information from $f_{l,t}^{app}$ according to the spatial carrier $f_{l,t}^{out}$.

Using these spatial correspondence weights A_{cross} , we calculate the local style statistics—mean M (color information) and standard deviation S (texture information)—from the appearance features $f_{l,t}^{app}$:

$$M = A_{cross} f_{l,t}^{app}, \quad (6)$$

$$S = \sqrt{A_{cross} (f_{l,t}^{app} \odot f_{l,t}^{app}) - (M \odot M)}. \quad (7)$$

Here, M is the attention-weighted local mean of the appearance features, and S is the corresponding local standard deviation, which captures the strength of texture variation to be transferred into the main branch. Finally, we blend this appearance onto the structurally correct output features. We first normalize the output features to remove their current style, and then re-apply the target style using the calculated statistics:

$$f_{l,t}^{out} \leftarrow S \odot \text{Norm}(f_{l,t}^{out}) + M. \quad (8)$$

Thus, M and S provide the local color and texture statistics, respectively, and the final update in Eq. 8 writes these appearance statistics back into the same $f_{l,t}^{out}$ feature that has already been structurally constrained by FTSC. This sequential application—first enforcing structure via replacement, then blending appearance via spatially attuned statistics transfer—enables effective disentanglement and control over the final synthesis I^{out} . Equation 8 therefore updates color and texture while keeping the FTSC-imposed spatial carrier intact.

C. Multimodal Prompt Modulator

Our ablation analysis shows that raw user prompts are often too concise to describe the semantic richness of RS scenes, especially when object counts, scene context, or local environmental cues need to be inferred from the condition map. To provide robust semantic guidance and reduce such mismatches between the user’s raw prompt ($\mathcal{P}$) and the structural condition (I^{struct}), our framework introduces the Multimodal Prompt Modulator (MPM). This pipeline programmatically generates a *Grounded Semantic Prompt* (GSP) by leveraging a Vision-Language Model (VLM), implemented with LLaVA in our full setting [20], to interpret both the user’s text prompt $\mathcal{P}$ and the structural condition map I^{struct} . This process operates through three stages (see Fig. 4): (1) the VLM first analyzes I^{struct} to extract a high-level semantic understanding of the scene layout; (2) it then undergoes semantic alignment, cross-referencing its interpretation with $\mathcal{P}$ and inferring critical details from I^{struct} , such as likely object instance counts; and (3) it synthesizes the final GSP under a protocol that mandates a satellite remote-sensing viewpoint and realism-enhancing keywords. The resulting GSP serves as a rich semantic prior used to generate an appearance reference image I^{app} (and corresponding features $f_{l,t}^{app}$), ultimately providing superior guidance for controlling appearance and complementing the structural control provided by FTSC.

D. Restart Refinement Strategy (RRS)

Our controlled generation stage already establishes the desired layout, but the late denoising regime can still leave subtle texture artifacts, color casts, or local shape distortions. These residual artifacts should not be interpreted as being introduced by FTSC alone. Native diffusion sampling may leave diffuse low-level noise or color inconsistency, while forceful structural replacement in FTSC can spatially anchor or amplify such residuals around condition boundaries. RRS is therefore introduced as a low-noise polishing stage for mixed-origin residual artifacts after structure-controlled generation, rather than as evidence that all artifacts are caused by FTSC. To address this and further elevate the photorealism of the final output I^{out} , we incorporate a final Restart Refinement Strategy (RRS). This strategy acts as a dedicated *polishing stage*, operating sequentially after the main structure-controlled generation phase guided by FTSC has largely concluded (at $t \leq \tau$).

Compared with the standard Stable Diffusion/DDIM sampling process, which follows a single monotonic reverse trajectory from high noise to low noise [8, 30], RRS adds a restricted

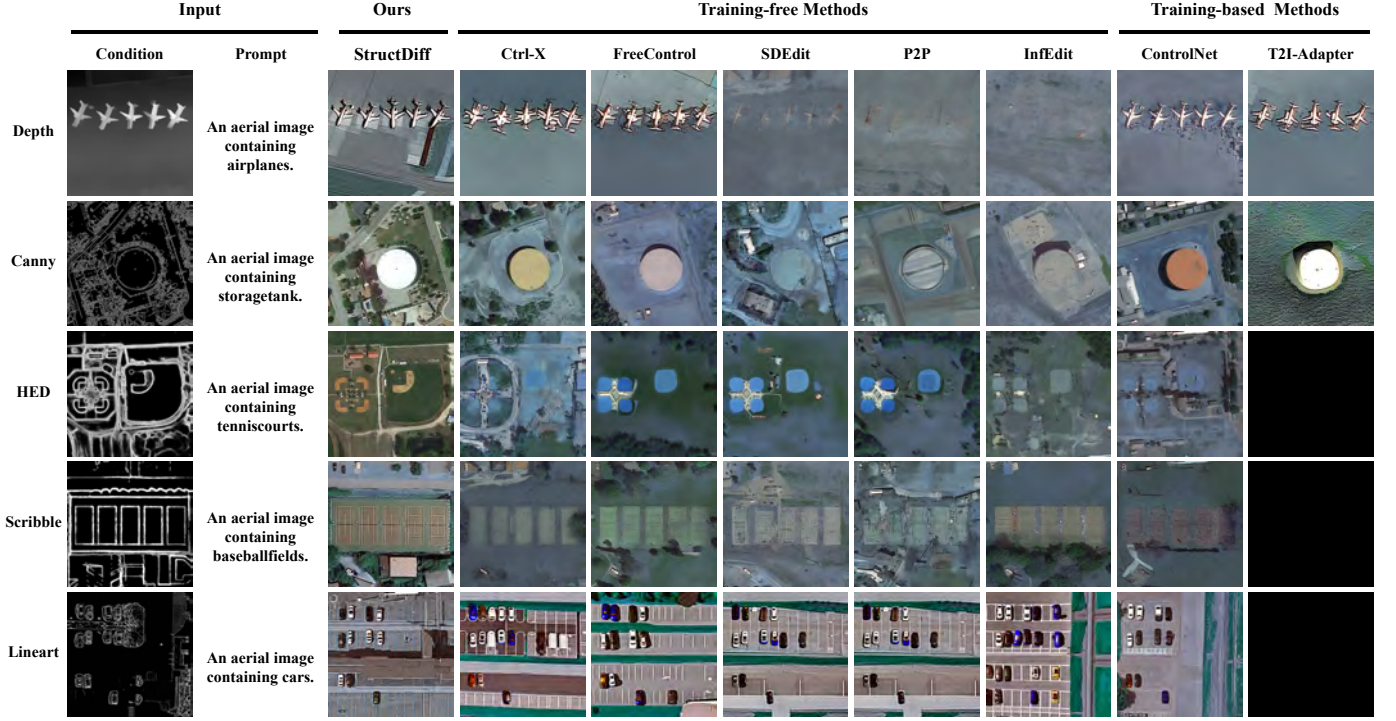


Fig. 5. Qualitative comparison with SOTA methods. StructDiff preserves structure and semantics more reliably than the training-free baselines. T2I-Adapter results are omitted for unsupported conditions.

local restart loop only after the structural layout has been established. It is also different from using restart sampling as a generic image-quality booster [22]: in StructDiff, the restart boundary is confined to a low-noise interval so that the sampler can re-denoise local artifacts while preserving the FTSC-constrained geospatial layout.

As depicted in Fig. 4, the RRS module takes the near-final latent state $x_{t_{min}}^{(0)}$ (corresponding to a low noise standard deviation σ_{min}) and performs N refinement cycles within a confined low-noise neighborhood. Each cycle $i \in \{0, \dots, N-1\}$ perturbs the latent back to a slightly higher noise level t_{max} (with $\sigma_{max} > \sigma_{min}$) and then denoises it back down. Specifically, the forward perturbation step applies noise via the diffusion forward kernel q :

$$x_{t_{max}}^{(i+1)} \sim q(x_{t_{max}} | x_{t_{min}}^{(i)}), \quad (9)$$

and a subsequent denoising step, employing the DDIM sampler $\mathcal{D}_{DDIM}$, refines the latent back towards t_{min} :

$$x_{t_{min}}^{(i+1)} = \mathcal{D}_{DDIM}(x_{t_{max}}^{(i+1)}, t_{max} \rightarrow t_{min}). \quad (10)$$

Equation 9 adds a controlled amount of noise to revisit local defects, and Eq. 10 denoises only within the restricted interval from t_{max} to t_{min} so the full high-noise trajectory is not reopened. This iterative loop of controlled noising (Eq. 9) and rediffusion (Eq. 10), inspired by restart sampling concepts, allows the model to correct subtle artifacts without disrupting the established geospatial accuracy, thereby enhancing the final perceptual quality of I^{out} .

IV. EXPERIMENTS

A. Experimental Setup

1) *Evaluation Protocol*: To create a stringent benchmark, we curated 1500 image-condition pairs from DIOR [31] across five structural guidance types: depth, Canny, HED, scribble, and lineart. We use a 10% held-out test split (150 pairs) and a 90% fine-tuning split (1350 pairs) for the training-based baselines (ControlNet [16], T2I-Adapter [24]). For fairness, ControlNet is fine-tuned and evaluated per condition family; the *Full Dataset* (5 conditions) row aggregates five condition-specific runs, whereas the *T2I-Adapter-supported* (2 conditions) block reports only the canny/depth subset supported by the official SDXL implementation. We also compare against representative training-free baselines, including Ctrl-X [25] and FreeControl [26], all on the SDXL 1.0 backbone [21]. We report **structural fidelity** (DINO Self-sim ↓) [32], **semantic fidelity** (CLIP Score ↑) [33], and **condition carry-over proxy** (Condition LPIPS ↑) [34]; higher LPIPS indicates weaker visual carry-over rather than a standalone validated cross-modality leakage measure.

The study consists of a primary DIOR benchmark, zero-shot DOTA transfer, blinded human evaluation, and a controlled downstream object-detection test; full details are provided in the supplementary material.

2) *Core Implementation*: Our framework is implemented in Diffusers [35] using the pretrained SDXL 1.0 backbone [21] with a DDIM sampler (50 steps). The key settings are $C = 600$, $\tau = 400$, $N = 3$, $[\sigma_{t_{min}}, \sigma_{t_{max}}] = [0.15, 0.35]$, and a LLaVA-v1.5 (7B) MPM [20]; the supplementary material reports the supporting FTSC and RRS sensitivity results. The

TABLE I

QUANTITATIVE COMPARISON ON DIOR. STRUCTDIFF GIVES THE STRONGEST SEMANTIC ALIGNMENT AND CARRY-OVER SUPPRESSION AMONG THE TRAINING-FREE METHODS WHILE REMAINING COMPETITIVE IN STRUCTURAL FIDELITY. BEST IN **BOLD**; SECOND BEST UNDERLINED.

Method	Full Dataset (5 conditions)			T2I-Adapter-supported (2 conditions)		
	Self-sim ↓	CLIP ↑	LPIPS ↑	Self-sim ↓	CLIP ↑	LPIPS ↑
<i>Training-Based Baselines</i>						
ControlNet (pretrained) [16] (CVPR, 2023)	0.138	0.288	0.655	0.141	0.290	0.648
ControlNet (fine-tuned) [16] (CVPR, 2023)	0.083	<u>0.312</u>	<u>0.711</u>	0.086	<u>0.315</u>	<u>0.708</u>
T2I-Adapter (pretrained) [24] (ICCV, 2023)	/	/	/	0.145	<u>0.282</u>	<u>0.670</u>
T2I-Adapter (fine-tuned) [24] (ICCV, 2023)	/	/	/	0.118	0.298	0.705
<i>Training-Free Baselines</i>						
SDEdit [36] (ICLR, 2022)	0.152	0.251	0.355	0.168	0.248	0.301
P2P [19] (SIGGRAPH, 2022)	0.191	0.239	0.281	0.203	0.242	0.255
InfEdit [37] (CVPR, 2024)	0.145	0.281	0.362	0.151	0.283	0.350
FreeControl [26] (ICLR, 2024)	0.129	0.296	0.672	0.133	0.301	0.659
Ctrl-X [25] (CVPR, 2024)	0.125	0.299	0.668	0.128	0.304	0.652
StructDiff (Ours)	<u>0.115</u>	0.318	0.715	<u>0.116</u>	0.321	0.712

VLM remains frozen and is used only at inference time, so it adds runtime and memory overhead without introducing any additional training stage. Remaining details, such as CFG, FTSC injection layers (l), and preprocessing, are deferred to the supplementary material. All experiments were conducted on NVIDIA A6000 GPUs.

B. Comparison with State-of-the-Art

1) *Quantitative Analysis*: Table I reveals the main trade-off: fine-tuned ControlNet achieves the best structural fidelity, while StructDiff achieves the highest CLIP and LPIPS scores. Given the modest CLIP gap over ControlNet (0.318 vs. 0.312), we interpret StructDiff as offering a better overall balance rather than exposing a categorical semantic failure of the training-based baseline. It also outperforms the *pretrained* versions of ControlNet and T2I-Adapter, highlighting the domain gap of natural-image pretrained controllers.

We do not include the plain SDXL backbone as a structural-control baseline because it has no mechanism for conditioning generation on an external structural input. The fair comparison is therefore among controllable methods operating on the same frozen backbone, such as Ctrl-X and FreeControl, together with training-based control models such as ControlNet and T2I-Adapter. Under this shared-backbone setting, the comparison is about control strategies rather than raw backbone strength.

2) *Qualitative Analysis*: Figure 5 shows that StructDiff remains structurally precise and semantically faithful. **Training-based** models like ControlNet remain structurally sound but can under-adapt semantics in some cases, while **training-free** baselines show the expected failures of leakage, structure loss, or low contrast. These observations are also consistent with the human study reported in the supplementary material.

C. Additional Validation

Beyond the primary DIOR benchmark, we use additional validation to test external generalization and perceptual quality. Detailed DOTA and human-evaluation results are reported in the supplementary material; in short, StructDiff remains the

TABLE II

QUANTITATIVE FTSC ABLATION. THE FIXED TIMESTEP $C = 600$ GIVES THE BEST AGGREGATE BALANCE ON THE CURRENT BENCHMARK.

Injection Strategy	Self-sim ↓	CLIP ↑	LPIPS ↑
Synchronous ($g(t) = t$)	0.128	0.305	0.675
Asynchronous ($C = 0$)	0.160	0.290	0.550
Asynchronous ($C = 200$)	0.135	0.310	0.650
Asynchronous ($C = 400$)	0.120	0.315	0.705
Asynchronous ($C = 600$)	0.115	0.318	0.715
Asynchronous ($C = 800$)	0.155	<u>0.320</u>	<u>0.710</u>

strongest zero-shot option on DOTA and receives the highest realism and structural-accuracy ratings in the blinded human study.

D. Ablation Study

We next isolate the three modules that address complementary failure modes.

1) *Ablation Study on Fixed-Timestep Structural Conditioner*: We compare the synchronous baseline ($g(t) = t$) with asynchronous strategies that cache features from early/high-noise ($C = 800$), middle ($C = 400, 600$), and late/low-noise ($C = 200$) denoising stages. Here we follow the reverse-diffusion convention used throughout the paper: larger numeric t means an earlier and noisier stage, while smaller numeric t means a later and cleaner stage. Figure 6 and Table II show that very small C values increase leakage, very large C values lose detail, and the middle-stage choice $C = 600$ gives the best aggregate balance, consistent with the per-condition results in the supplement.

2) *Ablation Study on Multimodal Prompt Modulator*: MPM addresses a different failure mode: even when structure is correct, concise user prompts can under-specify RS appearance.

We compare the FTSC+RRS baseline without MPM, a lightweight BLIP-2-based MPM, and the full LLaVA-based MPM while keeping FTSC fixed. Table III and Fig. 7 show that prompt enrichment improves semantic alignment and perceptual quality, while the BLIP-2 variant retains most of the gain at lower cost.

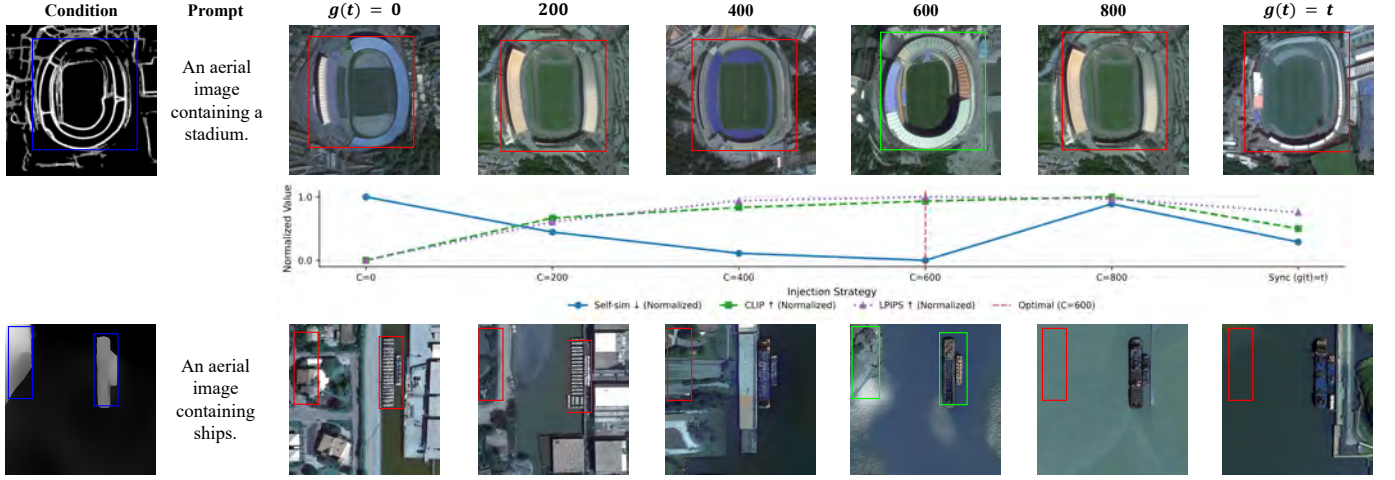


Fig. 6. FTSC ablation: synchronous guidance versus fixed-timestep caches. The middle-stage choice $C = 600$ provides the best structure-quality trade-off. MPM and RRS are enabled for all runs.

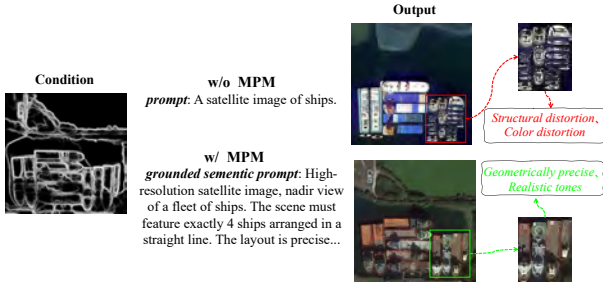


Fig. 7. Qualitative ablation for MPM. Prompt enrichment improves semantic alignment and reduces appearance errors.

TABLE III
MPM ABLATION WITH DIFFERENT PROMPT GENERATORS. THE BLIP-2 VARIANT RETAINS MOST OF THE GAIN AT LOWER MODEL COST.

Method	Self-sim ↓	CLIP ↑	LPIPS ↑
FTSC+RRS [w/o MPM]	0.117	0.295	0.701
FTSC+RRS+BLIP-2 MPM [Light]	0.116	0.311	0.709
FTSC+RRS+MPM [Full]	0.115	0.318	0.715

3) *Ablation Study on Restart Refinement Strategy*: RRS targets residual low-noise artifacts that remain after the main generation stage. To make the module progression explicit, we compare the bare FTSC baseline, the FTSC+MPM variant without RRS, and the full pipeline, and complement the main-paper comparison with supplementary sensitivity tests over the restart iteration count and refinement boundary. Figure 8 and Table V show that RRS removes local artifacts and improves LPIPS while keeping Self-sim and CLIP stable.

A more detailed source analysis of native sampling artifacts and FTSC-associated boundary artifacts is provided in the supplementary material.

E. Practical Validation

We next examine practical usability through low-data detection and inference efficiency.

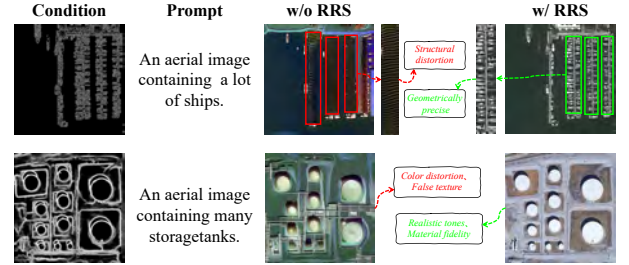


Fig. 8. Qualitative ablation of RRS. The refinement stage mitigates local appearance artifacts while preserving structure.

1) *Downstream Object Detection Evaluation*: We assess practical utility with a fixed YOLOv8-s detector on DIOR under 10%, 20%, and 50% low-data regimes, comparing real-only training against training augmented with StructDiff-generated images. Table IV shows consistent gains, especially in the most data-limited setting; the supplementary material summarizes the protocol.

2) *Inference Efficiency*: The detailed operating-point comparison for FTSC, MPM, and RRS is moved to the supplementary material. In brief, FTSC is the lightest structural-control backbone, while MPM and RRS trade extra runtime and memory for stronger semantic alignment and residual-artifact suppression.

V. CONCLUSION

In this work, we presented StructDiff, a training-free framework for controllable RS image synthesis that does not require task-specific retraining for new condition types. By combining fixed-timestep structural conditioning, grounded prompt modulation, and late-stage restart refinement, the proposed method provides a strong balance between structural fidelity, semantic alignment, and perceptual quality. Experiments on DIOR and DOTA, together with human evaluation and a documented low-data downstream detection protocol, further support its practical potential for controlled RS augmentation.

Looking ahead, we plan to explore more lightweight prompt-enhancement modules to reduce the dependency on

TABLE IV
DOWNSTREAM OBJECT DETECTION EVALUATION ON DIOR UNDER LOW-DATA REGIMES. HIGHER IS BETTER FOR BOTH METRICS.

Training Ratio	mAP@0.5			mAP@[0.5:0.95]		
	Real-only	Real + StructDiff	Gain	Real-only	Real + StructDiff	Gain
10%	41.3	45.8	+4.5	22.6	25.1	+2.5
20%	49.7	53.1	+3.4	28.4	30.5	+2.1
50%	60.8	62.7	+1.9	35.7	37.0	+1.3

TABLE V
RRS ABLATION. THE REFINEMENT STAGE IMPROVES PERCEPTUAL QUALITY WHILE KEEPING SELF-SIM AND CLIP STABLE; THE FTSC-ONLY ROW ISOLATES THE BARE STRUCTURAL BASELINE.

Method	Self-sim ↓	CLIP ↑	LPIPS ↑
FTSC (Base)	0.130	0.295	0.665
FTSC+MPM [w/o RRS]	0.116	0.317	0.682
FTSC+RRS+MPM [Full]	0.115	0.318	0.715

external large models. Furthermore, integrating our framework with domain-specific models for deep adaptation in areas such as digital cultural heritage, industrial design, and urban planning presents another key research direction. We believe this work offers valuable insights for controllable visual content creation and lays a solid foundation for future research.

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